

Using MOI Transformation Equations to Compute I_u , I_v , I_{uv} at some angle θ from x .
 Equations: Given or find for an Area (about its x and y axes): I_x , I_y , and I_{xy}
 (Note: If $I_{xy} = 0$, then this Area's I_x and I_y are principal axes (I_{MAX} and I_{MIN})).

To find I_u , I_v and I_{uv} about an arbitrary rotated coordinate system (at angle θ):

1. Equations in terms of θ .
 (Can use the I_u equation for both I_θ and $I_{\theta+90}$, just put the correct angle in the equation; so technically do not need the I_v eqn)

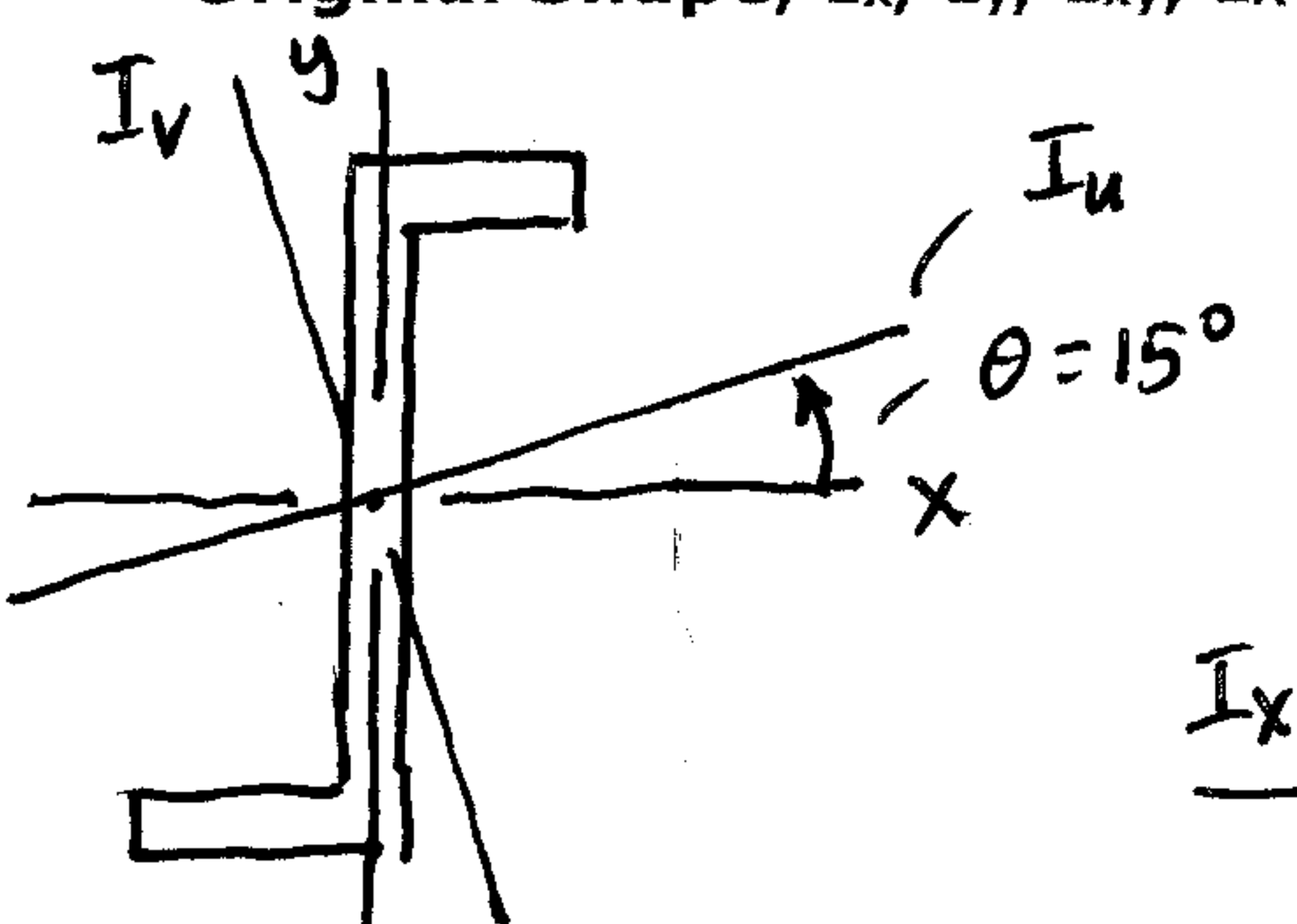
$$\begin{aligned} 1(a) \quad & I_u = I_x \cos^2 \theta + I_y \sin^2 \theta - 2I_{xy} \sin \theta \cos \theta \\ 1(b) \quad & I_v = I_x \sin^2 \theta + I_y \cos^2 \theta + 2I_{xy} \sin \theta \cos \theta \\ 1(c) \quad & I_{uv} = I_{xy}(\cos^2 \theta - \sin^2 \theta) + (I_x - I_y) \sin \theta \cos \theta \end{aligned}$$

For I_{uv} easier to use the 2θ form (2(c)) of the equation below.

2. Equations in terms of 2θ .
 (Slightly easier to calculate, no need to square sin and cos terms)

$$\begin{aligned} 2(a) \quad & I_u = \frac{I_x + I_y}{2} + \frac{I_x - I_y}{2} \cos 2\theta - I_{xy} \sin 2\theta \\ 2(b) \quad & I_v = \frac{I_x + I_y}{2} - \frac{I_x - I_y}{2} \cos 2\theta + I_{xy} \sin 2\theta \\ 2(c) \quad & I_{uv} = \frac{I_x - I_y}{2} \sin 2\theta + I_{xy} \cos 2\theta \end{aligned}$$

Original Shape, I_x , I_y , I_{xy} , I_{AVG} , $(I_x - I_y)/2$, $I_x = 15.7 \text{ in}^4$; $I_y = 1.04 \text{ in}^4$
 $I_{xy} = 2.68 \text{ in}^4$
 $I_{AVG} = \frac{I_x + I_y}{2} = \frac{15.7 + 1.04}{2} = 8.39 \text{ in}^4$
 $\frac{I_x - I_y}{2} = \frac{15.7 - 1.04}{2} = 7.35 \text{ in}^4$



Using the θ equations to find I_u , I_v , I_{uv} at $\theta = 15^\circ$.

$$I_u = I_{15} = 15.7 \cos^2(15) + 1.04 \sin^2(15) - 2(2.68) \sin 15 \cos 15$$

$$I_u = I_{15} = 13.4 \text{ in}^4$$

$$I_{uv} \text{ (at } \theta=15^\circ) = 7.35 \sin 30 + 2.68 \cos 30 \Rightarrow I_{uv} = 6.0 \text{ in}^4$$

(use Egn 2(c))
Easier!

$\theta = 105^\circ$
for I_v

$$I_v = I_{105} = 15.7 \cos^2(105) + 1.04 \sin^2(105) - 2(2.68) \sin 105 \cos 105$$

$$I_v = 3.36 \text{ in}^4$$

$$\text{Egn 1(b), } \theta = 15^\circ: I_v = 3.36 \text{ in}^4$$

For I_v , can use
 Egn 1(b) with $\theta = 15^\circ$,
 OR can use Egn 1(a)
 with $\theta = 90 + 15$
 $= 105^\circ$

For I_{uv} easier to use the 2θ form (2(c)) of the equation below.

2. Equations in terms of 2θ .

(Slightly easier to calculate, no need to square sin and cos terms)

2(a)

$$I_u = \frac{I_x + I_y}{2} + \frac{I_x - I_y}{2} \cos 2\theta - I_{xy} \sin 2\theta$$

2(b)

$$I_v = \frac{I_x + I_y}{2} - \frac{I_x - I_y}{2} \cos 2\theta + I_{xy} \sin 2\theta$$

2(c)

$$I_{uv} = \frac{I_x - I_y}{2} \sin 2\theta + I_{xy} \cos 2\theta$$

Using the 2θ equations to find I_u , I_v , I_{uv} at $\theta = 15^\circ$, $2\theta = 30^\circ$.

These are slightly easier to calculate because do not have to eval $(\cos \theta)^2$ and $(\sin \theta)^2$...

Remember: We already calculated $I_{x0} = 8.39 \text{ in}^4$

$$\left(\frac{I_x - I_y}{2} \right) = 7.35 \text{ in}^4$$

$$I_u = I_{15} = 8.39 + 7.35 \cos 30 - 2.68 \sin 30$$

$$I_u = I_{15} = 13.4 \text{ in}^4$$

$$I_v = 8.39 - 7.35 \cos 30 + 2.68 \sin 30$$

$$I_v = 3.36 \text{ in}^4$$

$$I_{uv} = 7.35 \sin 30 + 2.68 \cos 30$$

$$I_{uv} = 6.0 \text{ in}^4$$